Hybrid Application Development and Implementation

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***Abstract*—Innovation of powerful-smaller chips, processors, etc. has increased the ability of mobile device's execution capabilities manifold. Taking into consideration the applications which doesn't require much computational and hardware resources and in turn rely heavily on basic functionalities and server-side computations. Such applications also require deployment on multiple-different platforms (OS). Ignoring the execution performance aspects for a minute, how can we reduce the development time/cost for such applications and optimize the rather brute-forced approach of multiple platform deployments? A control interface application for various Internet of Things (IoT) may falls under this category and with the boom of IoT, the requirement to solve the above question becomes vital. Hybrid Mobile Applications are applications that can be deployed on various platforms with a single codebase. The codebase is written once in HTML5, CSS, and JavaScript (or Typescript) for the functionality of the application. This codebase is then wrapped in native-specific containers that can be deployed on multiple platforms. Providing a 'code-once-deploy-anywhere' model, Hybrid Applications offers an optimal solution to the application type mentioned above. In this paper we venture the world of Hybrid Mobile Application and its implementations along with the optimization of the same.**

Keywords— mobile application development, cross-platform, cordova, hybrid applications, ionic

I. INTRODUCTION

Development of portable devices has fueled the need of Internet of Things (IoT) which in turn has fueled the development of smart devices requiring control applications. These applications (deployed on mobile), having basic functionalities doesn't require much resources and hardware support. Although, basic in nature these control applications have a requirement i.e.- 'compatibility on different platforms'. This has motivated the research direction and development of Hybrid applications. Hybrid Application development facilitates cross-platform code sharing due to its 'code-once-deploy anywhere' model unlike its adversary the native applications [8]. Keeping these control applications aside, any mobile application which aims to achieve cross-platform code sharing can be benefited by Hybrid Applications. At the same time the requirement of platform-specialist developers and cost to company decreases considerably. Fig. 1 Charts out the Capability vs Platform attribute for different types of mobile applications.

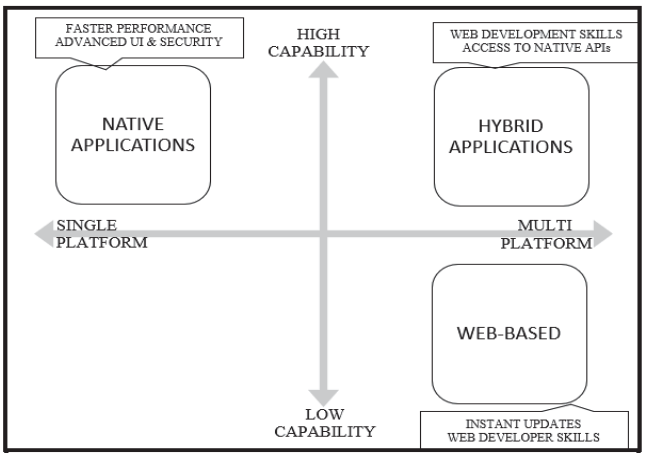


Fig. 1. Capability vs. Platform graph

A. What is Cross Platform Development?

Cross Platform Application Development refers to creation and development of Mobile applications to work on different Platforms (namely Android, iOS, Windows.). We can achieve cross platform development in two ways:

1) Bruteforce Multi-Platform Native Application: Developing different native Applications for their respective platforms.

2) Hybrid Application Development: Developing a singular codebase that achieves application functionality and

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exporting this singular codebase in different applications compatible to different platforms. This method of development inspires Hybrid application development.

We formally define Native and Hybrid applications as:

1) Native Applications: Native mobile applications are applications developed with the SDK (Software Development Kit) in platform-specific programming language. These applications are limited to a single platform and have the highest performance and response. [1]

2) Hybrid Applications : Hybrid mobile applications are applications in which codebase once created can be deployed on any platform by wrapping the codebase in platform specific wrapper. These applications once coded can be deployed on different platforms with just different export command parameters [1]. The comparison (Table I) clearly represents some important aspects such as listed below:

- Native applications perform better as compared to hybrid applications.
- Hybrid applications uses an amalgamation of much simpler coding languages (HTML5, CSS, JS) as compared to different coding languages of native applications [2,4,15].
- Hybrid applications promotes cross-platform code sharing.

II. Motivation

Earlier whenever there was a need of cross platform applications companies were forced to hire different expert developers or group for creating applications compatible to a particular platform. This resulted in high cost to company, a brute forced approach towards cross platform applications, and a prolonged development cycle. Fig. 2 depicts a generic situation for cross platform mobile development [3].

TABLE I. Comparison between native & hybrid applications

Native Mobile Application	Hybrid Mobile Application
Developed using software development kit and platform specific programming language.	Codebase (written in a particular language) is enclosed within a platform specific-native container.
Hinders deployment of an application on multiple mobile platforms.	Encourages cross platform mobile application deployment.
The application is written according to the mobile platform.	Uses HTML5, CSS3 and JavaScript for different parts of the application.
Hardware related performance is high [3].	Hardware performance is low [3].
Suitable for Games, Audio-Video Editors.	Suitable for web based mobile application having client-server relation.
Prominent Tools & Libraries Include : For android : Android Studio, Android SDK, Java, Eclipse For iOS : Xcode	Some of the tools & libraries for Hybrid application development are: Ionic Cordova, Titanium Appcelator, etc. [4]
E.g.: Asphalt 8, Clash of Clans	E.g.: Instagram, Uber

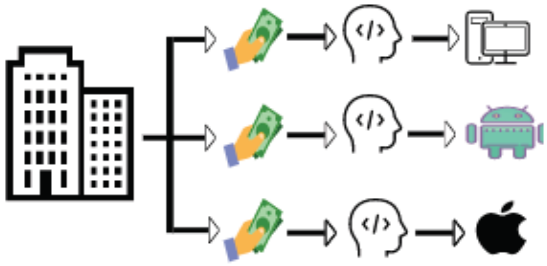


Fig. 2. Generic approach to achieve cross-platform mobile applications.

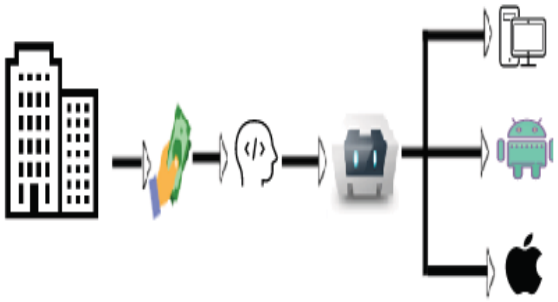


Fig. 3. Using Cordova to achieve cross-platform mobile applications.

With Hybrid Applications the company can hire a single developer or a group whose sole work is to develop the application once followed by Cordova exports to different platforms. Cordova will wrap the code in native containers with respect to the platform mentioned in the export command. Fig. 3. depicts Cordova approach to achieve cross-platform applications.

III. Problem Statement

The current research aims to develop and implement a hybrid application for mobile users. Table II presents various challenges and their corresponding solutions

TABLE II. Challenges & Corresponding Solutions

S No.	Challenge	Proposed Solution
1	How to achieve "Code-once-deploy-anywhere" model for mobile application?	We've achieved the "code-once-deploy anywhere" model for mobile application with Apache Cordova Framework and Ionic 3 [4]. Apache Cordova framework and Ionic 3 wraps the singular codebase in a wrapper code, this wrapper code is native to its respective platform [15].
2	"How to implement Hybrid-Application?"	Implementation of Hybrid Applications is done on the Apache Cordova framework and Ionic 3. The code is written in

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		JavaScript/TypeScript (with special APIs to access device resources), HTML and CSS [2, 4, 15]. The respective documentations [11, 12, 13] have been referred and online resources to familiarize with Ionic Framework, CLI commands (Cordova) and Angular JS.
3	Which database to choose?	We have arrived at the decision to use MySQL database due to the vast amount of research that have been done on the MySQL database [5, 14]. It also gives us an additional option to implement security on the backend side. NoSQL lacks standardization and reporting tools for analysis.
4	How to handle security of Hybrid Application?	We can achieve security improvements with extra layers of security measures on the backend side of the application (as mentioned in research challenge also on the frontend of the application with emphasis on improved security for plugins that access the device's hardware. [6, 7, 9]
5	How to increase performance of Hybrid Application?	Talking about the performance metrics of the hybrid applications and native application. Hybrid Applications performs poorly as compared to native application. Plug-in managements can result in increased performance. Optimization of HTML modules along with assets can result in increased performance [16].

IV. Hybrid Application Development: Challenges

A. Security Challenges

In [7, 9, 10] authors have mentioned various security loopholes for Ionic Cordova applications. Testbed screening, malicious code injection, cross-site scripting were intentionally performed on hybrid applications to list out various factors affecting security. Major contributions of the same are caused by:

1) JavaScript: JavaScript scripts are easier target to code injections, cross-site scripting [7]. Being an integral part of ionic hybrid application, they possess a greater threat unlike HTML applications as the browser running the JS are not sandboxed and are directly connected to phones resources.

2) Community Plugins: A plugin is a bit of add-on code that provides JavaScript interface to native components. They allow hybrid app to use native device capabilities beyond what is available to pure web apps [6]. A malicious intent plugin can exploit almost every component of the device.

B. Performance Challenges

In [1] the author discusses briefly about the metrics for performance through his tests, the metrics are as follows:

1) CPU Utilization: The CPU utilization for Native applications is far less than that of Hybrid application. This is because native applications are developed with highest efficiency to achieve maximum optimization with the help of SDK's specially written for its architecture and framework.

2) RAM Utilization: Mobile applications run on small physical devices and that is why usage of RAM represents a challenge.

3) Development Time/Cost Factor: Developing applications for each platform or platform version would introduce additional requirements for designers and developers. If the application supports three mobile platforms, it causes a tripling effort required to develop applications. The results show that the development of hybrid applications requires less effort, which is expected outcome of this testing.

V. Solution Approach

A. Proprosed Solution

The Implementation of Hybrid mobile applications, with emphasis on security and performance raises serious confrontation for solutions, we came up with the following idea as depicted in Fig. 4. that we can follow to achieve an acceptable result.

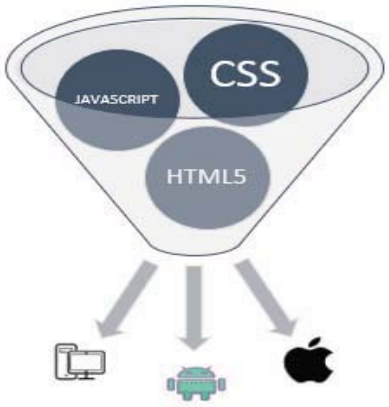


Fig. 4. Proposed Solution.

B. Security Improvement

1) Securing JavaScript: We can secure JavaScript scripts by obfuscating; using an obfuscator will minify scripts readability thus reducing it from analyzing and theft. Still reverse engineering becomes a problem [7].

2) Reducing Cross-site scripts to trigger: Using whitelist (a firewall like mechanism to filter external http/https requests) will minimize the code's vulnerability towards cross-site script attacks from unknown servers [7].

3) Remote JavaScript Execution: Executing some JavaScript files on a remote desktop will reduce the risk of injection attacks where a single line of code can wreak havoc on device components [7].

4) Access Control for community plug-ins: Developing an Access control for community plugins will filter malicious

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plugins and give full authority to user on execution or termination of community plugins [6].

5) Role Based Access for CRUDs: Limiting CRUD operations as per the roles involve, limiting access of certain roles for certain operations on certain data tables or databases [5].

C. Performance Improvements

1) Minifying/Obfuscating Javascript: We can secure JavaScript scripts by obfuscating; using an obfuscator will minify the size of the scripts while removing unused packages [16].

2) Native Scrolling: Using Native Scrolling to improve the application scroll render will reduce the load on hardware components [16].

3) Ahead-of-Time Compilation : Using Angular's ahead-of-time compiling compiles the scripts during build time instead of its adversary i.e. Just-in-time compiling which compiles app during run-time [11, 12].

D. Architecture

Fig. 5. shows the architecture explaining the working of a hybrid application interacting with its database, codebase, and phone hardware [2]. As mentioned, Hybrid applications uses JavaScript, HTML5 and CSS for its codebase, reverse engineering the same becomes much easier and hence it's easier to inject the codebase with malicious codes. Additionally as Hybrid application uses plug-ins to interact with device component, a malicious plug-in can exploit device's component.

Fig. 6. shows the redefined solution-approach architecture. This architecture when applied with the minified/obfuscated code can prove difficult to reverse engineer, while adding extra layers of protection for the essential components and adds positively to the performance aspects as a minified-optimized UI/UX code reduce the performance costs.

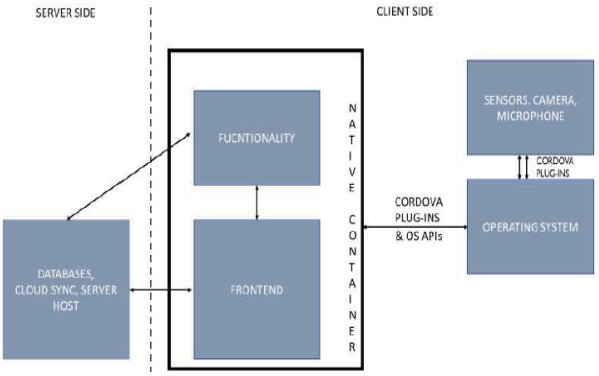


Fig. 5. Architecture diagram of an hybrid application [2].

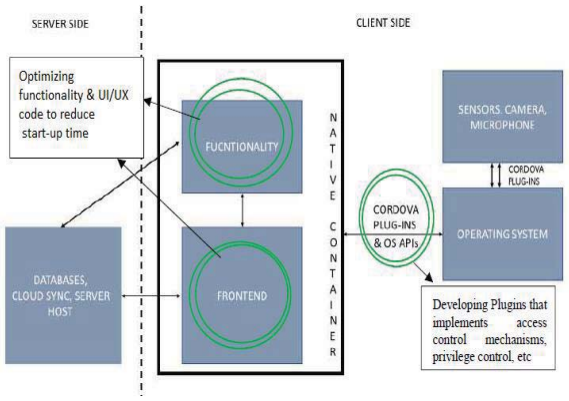


Fig. 6. Architecture diagram of the proposed solution

VI. Implementation

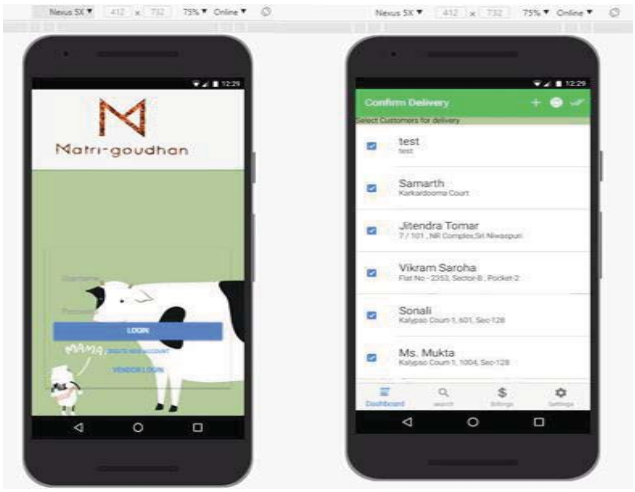


Fig. 7. Sample Application's Login & Vendor distribution list page.

A. Development of Hybid Mobile Application.

The sample Hybrid mobile application we created, intends to simplify and digitalize the milk distribution system for a milk distributor. As this application requires little to none hardware resources this is a perfect use-case application to implement the hybrid application model. Fig. 7. demonstrates a screenshot of the result application. The intended end users are:

1) Admin/Vendor: The admin/vendor user will use the application to get a list of all the deliveries (i.e. customers and respective quantities) he has to perform on a day to day basis. The application will also provide this user to view a summarized report of all the customer (their milk requirement, payment) to facilitate and achieve monetary transactions. This user will also have the facility to add a customer for the service (add customer to database).

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2) Customer: The customer user will use the application to register himself for the milk distribution service (addition of a new customer), update his/her milk requirement as many times he wants, view their outstanding amount with a payment directive to facilitate monetary transactions. The user can also view his previous transactions. These roles will have Role Based Access-Control dynamics for database access to improve security [5].

B. Functional & Non-Functional requirements.

The functional and non-functional requirement for the sample hybrid application are as follows:

1) Functional requirements:

a) Separate login pages for different end users.

b) Facilitate monetary transactions.

c) View past transactions (for customer end user).

d) Display daily delivery list and customer data (for admin user).

2) Non-Functional requirements:

a) Should have a backend server component to facilitate data sharing/availability through CRUDs.

b) Should have a cloud deployed columns.

C. Backend Components

To satisfy the functional and non functional requirements a backend server side has been hosted at the respective URL from 000webhost.com. The server facilitates the data traversal request from the server side to client side using PHP scripts & MySQL database (which are also hosted at the localhost of the same server) and takes care of the CRUDs, data availability and cloud storage.

Up to 15 PHP files loaded with SQL queries are hosted to satisfy the POST/GET request from the front-end as shown in Fig.8.

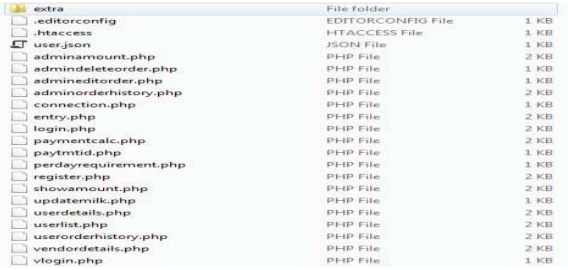


Fig. 8. Server's files, consisting majorly PHP scripts.

Additionally, A single database has been deployed with 7 tables that contains the data for the sample application to facilitate user orders, payments etc. These data are accessed by PHP files. Fig. 9. depicts the schema for the sample application's database.

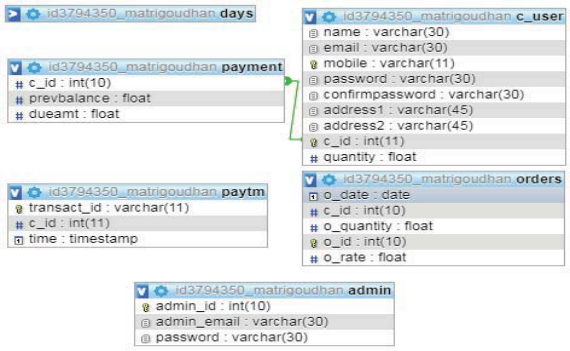


Fig. 9. Sample App's database schema.

D. Frontend Components

The developed sample application's code is coded in HTML5, CSS, and TypeScript to achieve its hybrid functionality, Fig. 10.

1) HTML5: For UI/UX purposes.

2) CSS (.scss): For UI/US purposes.

3) Typescript: To achieve achieve logical functionality and for HTTP POST/GET responses.

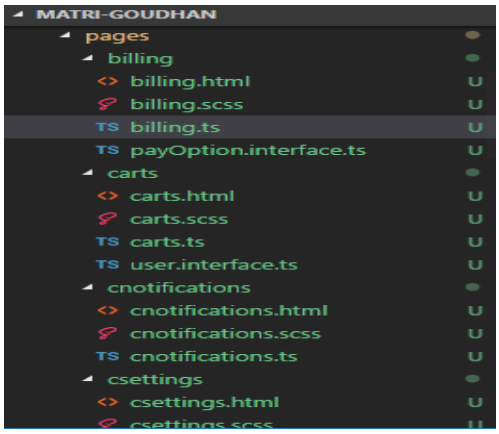


Fig. 10. Frontend files.

VII. Conclusion And Future Scope

The study summarizes recent development in the field of Hybrid Applications. Implementation techniques along with architecture of the technologies involved are discussed accordingly. We created and implemented a hybrid application — The sample application, whose architectural overview is directed in Fig.11. One of the major research area for future must be improvement & optimization of performance metrics in hybrid applications. Another area for research inspiration lies for tighter security to safeguard data as well as the hardware of the device.

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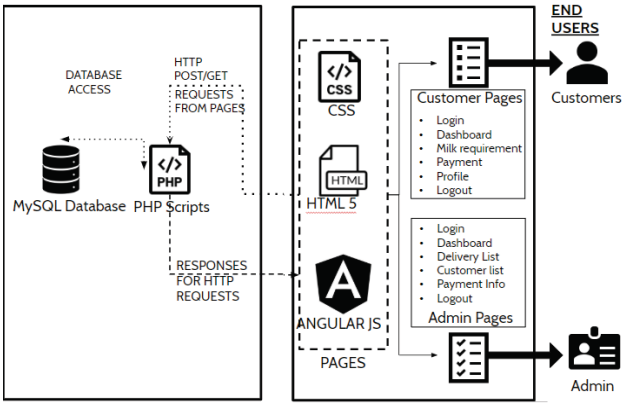


Fig. 11. Architectural overview of the developed sample application.

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